

2.1.1. List operations

57:42

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- **Add:** Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display:** Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit:** Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).

Code:

```
a = []
```

```
def Add(x):
```

```
    a.append(x)
```

```
    print(f"List after adding: {a}")
```

```
def Remove(x):
```

```
    for l in a:
```

```
        if (x==l):
```

```
            found = True
```

```
            break
```

```
        else:
```

```
            found = False
```

```
    if(found):
```

```

        a.remove(x)

        print(f"List after removing: {a}")

    else:

        print("Element not found")

def Display():

    if(len(a)==0):

        print("List is empty")

    else:

        print(a)

a = list()

for i in range(1,100):

    print("1. Add\n2. Remove\n3. Display\n4. Quit")

    Choice=int(input("Enter choice: "))

    if(Choice==1):

        x=int(input("Integer: "))

        Add(x)

    elif(Choice==2):

        if(len(a)==0):

            print("List is empty")

        else:

            x=int(input("Integer: "))

            Remove(x)

```

```

elif(Choice==3):

    Display()

elif(Choice==4):

    break

else:

    print("Invalid choice")

```

Explorer

Average time

0.262 s

261.67 ms

Maximum time

0.436 s

436.00 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1

165 ms

Debug

Expected output	Actual output
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 11	Integer: 11
List after adding: [11]	List after adding: [11]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 25	Integer: 25
List after adding: [11, 25]	List after adding: [11, 25]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 2	Enter choice: 2
Integer: 26	Integer: 26
Element not found	Element not found
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 2	Enter choice: 2
Integer: 11	Integer: 11

Activate Windows

Go to Settings to activate Windows.